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PAPER

Origin of regularity in different nuclei from the viewpoint of the effective quadrupole deformation and triaxiality

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E-mail: a.hosseinneshad@tabrizu.ac.ir**Keywords:** regular nuclei, effective quadrupole deformation, interacting boson model (IBM), random matrix theory (RMT)RECEIVED
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22 May 2024**Abstract**

This paper presents a theoretical investigation of the possible relation between the regularity of nuclei and the quadrupole deformation of different levels (and also triaxiality). The present paper aims to uncover the underlying physical reasons for this regularity. To this aim, we calculated the effective quadrupole deformation of different levels using the interacting boson model. Also, different ratios between the quadrupole deformations of the energy levels in the ground, beta, and gamma bands are defined. The results for the energy levels confirm the correctness of labeling these states by the quantum numbers of U(5) and SU(3) dynamical symmetry limits (and also the extraction processes). Also, we observed a repetition pattern for these ratios for regular nuclei. Of course, the regularity and sameness of repetition patterns for the levels of the rotational bands are more than the levels of the ground band. For further study, we analyzed the effective quadrupole deformation values of different levels of regular nuclei using random matrix theory. The results show a strong statistical correlation for these quantities and confirm the observed repetition pattern. Also, the results of our studies showed that regular nuclei have triaxial properties.

1. Introduction

The presence of deformation and similar structures in nuclear systems causes some nuclei to be classified as regular nuclei [1, 2]. The statistical study of the structure of nuclei is possible using the random matrix theory (RMT) [3–8]. Alhassid and Whelan in [9] (by using the RMT) introduced a regular region in the symmetry triangle [10] of the interacting boson model (IBM) [11]. The mentioned region is called the Alhassid-Whelan arc of regularity. According to the results of [9], the spectral statistics of the nuclei related to the ‘Alhassid-Whelan arc of regularity’ region are similar to the Gaussian distribution. There is a correlation and regularity between the data according to the Gaussian distribution [3, 12]. L. Amon and R. F. Casten (in their studies) showed that there is a specific repetition pattern for the energy levels of the nuclei located in the ‘Alhassid-Whelan arc of regularity’ region [13, 14]. In their studies, L. Amon and R. F. Casten proved the existence of a repetition pattern for the energy levels (for the nuclei located in the ‘Alhassid-Whelan arc of regularity’ region). We found new repetition patterns for regular nuclei (based on their experimental quadrupole transition probabilities) in [2]. We also introduced new nuclei as new candidates for regularity.

The shape is one of the features that can be attributed to describe systems. The most common deformation mode is the quadrupole deformation one in nuclear systems [15–18]. The quadrupole deformation resulting from the quadrupole interaction is caused by the transition of nucleons between energy levels (which was studied by Kumar in 1974 [19]). Also, D. Cline *et al* studied this issue more widely in their articles [20, 21]. Using the IBM, we calculated the effective quadrupole deformation of different levels (for the respective isotopes) in [22, 23].

For the present study, we first need to know the essential background details about regular nuclei, quadrupole deformation, IBM, and RMT. ‘Regular nuclei’ have a stable and well-defined structure. These nuclei have a characteristic arrangement of protons and neutrons, resulting in a specific number of nucleons (the

collective term for protons and neutrons). ‘Regular nuclei’ are typically found in stable isotopes of elements and follow repetition patterns in their energy levels and properties. Quadrupole deformation is a phenomenon observed in certain atomic nuclei where the shape of the nucleus deviates from a perfect sphere. In these nuclei, the distribution of protons and neutrons is not symmetric, resulting in an elongated or flattened shape. This deformation is characterized by a non-zero quadrupole moment (this quantifies the deviation from a spherical shape). In understanding nuclear structure and properties (such as energy levels, electromagnetic transitions, and nuclear reactions), quadrupole deformation plays a crucial role. The Interacting Boson Model (IBM) is a theoretical framework used to describe the behavior of atomic nuclei based on the concept of collective motion. According to this model, nucleons in the nucleus can be treated as interacting bosons (these are particles with integer spin). IBM assumes that pairs of nucleons act as collective bosons (either in a symmetric or mixed-symmetric configuration). This gives rise to different types of collective excitations. The IBM can explain various nuclear phenomena, such as rotational bands and vibrational states, particularly in medium-mass and heavy nuclei. Random Matrix Theory (RMT) is a mathematical framework that originated in the quantum physics field but has found applications in various disciplines, including nuclear physics. RMT deals with the statistical properties of matrices with random elements. In nuclear physics, RMT is used to study the statistical behavior of energy levels in atomic nuclei. It provides a theoretical basis for understanding the complex patterns observed in nuclear spectra (particularly in systems with chaotic or random behavior). RMT has been successful in explaining various statistical properties of energy levels. It has contributed to our understanding of quantum chaos in nuclei.

There are several reasons why regular nuclei and their quadrupole deformation are significant in nuclear physics. Firstly, regular nuclei are those that do not exhibit exotic properties such as high spin or isomeric states, making them representative of the majority of atomic nuclei. Understanding their properties provides a foundation for studying more complex systems. Quadrupole deformation refers to the deviation of a nucleus from a perfectly spherical shape, which is caused by the distribution of protons and neutrons within the nucleus. This deformation affects various nuclear properties, such as energy levels, electromagnetic transitions, and decay rates. Studying the quadrupole deformation of regular nuclei helps to understand the underlying nuclear structure and dynamics. Furthermore, by providing insights into the collective behavior of nucleons, the study of ‘regular nuclei’ with quadrupole deformation contributes to the existing body of knowledge in nuclear physics. Collective motion arises due to the interaction between nucleons and can be described by models such as the collective Bohr-Mottelson model or the interacting boson model. These models have successfully explained various nuclear phenomena, and studying regular nuclei with quadrupole deformation helps refine and validate them. The current state of research in this area has made significant progress in understanding the properties of ‘regular nuclei’ with quadrupole deformation. Experimental techniques (such as gamma-ray spectroscopy and scattering experiments) have provided valuable data on energy levels, transition probabilities, and other nuclear observables. Theoretical models have also been developed to describe the collective behavior of these nuclei. However, there are still gaps in our understanding that this paper aims to address. The precise determination of nuclear deformation parameters for specific isotopes is one such gap. While some nuclei have been well-studied, there are still many isotopes for which experimental data is limited or lacking. This paper aims to contribute to filling these gaps by providing new results and theoretical analysis for a set of regular nuclei with quadrupole deformation.

In this article, we aim to study the repetition patterns in regular nuclei (as well as new nuclei introduced as candidates for regular nuclei) [2] based on the study of effective quadrupole deformations of different levels. The existence of a repetition pattern for two observables (energy levels [13, 14] and quadrupole transition probabilities [2]) proved. Also, in [2], we proved that the existence of a similar structure leads to the creation of regularity in the respective nuclei. In the present article, we will show that similar deformations cause these nuclei to be classified as regular nuclei. In this regard, we use IBM to calculate effective quadrupole deformations. Also, for further study, we will study the statistical distribution of the effective quadrupole deformations for our target levels using RMT.

2. Methods and results

In comparison with previous studies, the present article lies in its approach to understanding the repetition pattern of energy levels in regular nuclei. By considering the effective quadrupole deformation and triaxiality, the paper aims to uncover the underlying physical reasons for this pattern. ‘Regular nuclei’ are of great importance in nuclear physics because they exhibit a repeating pattern of energy levels (which allows for the various theoretical models and techniques applications). The study of ‘regular nuclei’ helps to deepen our understanding of nuclear structure and dynamics and provides insights into the fundamental properties of atomic nuclei. One key question in this field is why given nuclei exhibit regularity in their energy levels while

others do not. Understanding the origin of this regularity is crucial for developing a comprehensive understanding of nuclear structure and the underlying interactions between nucleons. Studying regular nuclei can also help researchers refine and improve theoretical models used to describe nuclear phenomena. The repetition pattern observed in regular nuclei has implications for other nuclear properties and phenomena. For example, it can reveal nuclear shapes (and deformations) and the collective motion of nucleons inside the nucleus. This knowledge is essential for predicting and interpreting various nuclear reactions and processes. Furthermore, the regularity in energy levels of regular nuclei may have practical applications in technology and energy production. For instance, it can be utilized in designing and optimizing nuclear reactors, where a thorough understanding of nuclear structure and behavior is essential for safe and efficient operation.

The study of nuclear structure and related phenomena is performed using experimental and theoretical observables [8, 24–28]. There are various theoretical models for studying nuclear systems [29]. IBM [30–33] is one of the best models for describing the nuclei features and has been used in many studies, including [5, 34–43]. We choose the Interacting Boson Model (IBM) as the preferred theoretical model for calculating the quadrupole deformation of energy levels for several reasons. Firstly, IBM successfully described the collective properties of nuclei, including their deformation [44–47]. Based on the collective motion concept, nucleons (protons and neutrons) behave like bosons, interacting through effective forces. This model has been widely used and has provided accurate predictions for various nuclear properties [48–50]. Secondly, IBM incorporates the symmetries and interactions relevant to the quadrupole deformation of nuclei [51–54]. It considers the collective motion associated with the quadrupole shape of the nucleus, which is characterized by elongation or compression along the principal axes. By including these interactions, IBM can accurately describe the energy levels and their fluctuations in deformed nuclei. Furthermore, the IBM allows for a systematic description of nuclear structure across different regions of the nuclear chart. It provides a framework for understanding the evolution of nuclear properties as a function of proton and neutron numbers. This makes it a valuable tool for studying the quadrupole deformation of energy levels in a wide range of nuclei. Lastly, IBM has been extensively tested against experimental data and has shown good agreement in reproducing the observed energy levels and their fluctuations in deformed nuclei. This confirms its reliability and suitability for calculating the quadrupole deformation of energy levels.

The IBM was introduced (by A. Arima and F. Iachello) to describe the collective properties of semi-heavy and heavy even–even nuclei [30–33]. In this model, each pair of nucleons is considered a boson. Each boson is equal to half of the protons and neutrons outside the closed shell (corresponding to magic numbers). The boson number corresponds to protons or neutrons counted from the nearest closed shell. For example, if there are 76 neutrons, the relevant number of bosons is $(82-76)/2 = 3$ and not $(76-50)/2 = 13$. The boson numbers of protons and neutrons are denoted by N_π and N_ν , respectively. Thus $N_B = N_\pi + N_\nu$ represents the boson number for the nucleus. The IBM-1 version does not differentiate between N_π and N_ν , s and d-bosons have spin-parity $J^\pi = 0^+$ and $J^\pi = 2^+$, respectively. The s and d-boson creation and annihilation operators make operators related to nuclear observables. The general Hamiltonian [31, 55] of IBM is according to relation

$$\hat{H} = c \left[(1 - \eta) \hat{n}_d - \frac{\eta}{4N_B} \hat{Q}^\chi \cdot \hat{Q}^\chi \right]. \quad (1)$$

In relation 1, the quantity N_B is the boson number, and $\hat{Q}^\chi = (\hat{d}^\dagger \hat{s} + \hat{s}^\dagger \hat{d})^{(2)} + \chi (\hat{d}^\dagger \hat{d})^{(2)}$ is the quadrupole moment operator. Also, $\hat{n}_d = \hat{d}^\dagger \cdot \hat{d}$ is equal to the number of d-bosons, and variables η & χ represent Hamiltonian control parameters. The control parameters in the three dynamic limits of IBM are as follows:

$$\begin{cases} \text{U(5): } \eta = 0 \text{ and } \chi = 0 \\ \text{O(6): } \eta = 1 \text{ and } \chi = 0 \\ \text{U(3): } \eta = S = 1 \text{ and } \chi = -\sqrt{7/2} \end{cases}$$

To calculate the effective quadrupole deformation, we must first obtain quadrupole shape constants q_2 according to the relation [56]

$$q_2(L_n^+) = e^2 \langle L_n^+ | \hat{Q}^\chi \cdot \hat{Q}^\chi | L_n^+ \rangle. \quad (2)$$

In equation (2), the eigenfunction $|L_n^+\rangle$ in the form of IBM in three dynamic limits U(5) (for spherical nuclei), O(6) (for γ -unstable nuclei), and SU(3) (for deformed nuclei) are respectively as $|n_d, n_s, \tau, L\rangle$, $|n_d, n_s, \tau, \sigma, L\rangle$, and $|n_d, n_s, K, (\lambda, \mu), L\rangle$. How the creation and annihilation operators on eigenfunctions can be found in [11, 22, 23].

In deformed nuclei due to the existence of degeneracy in beta and gamma bands, other methods (such as partial dynamical symmetry (PDS)) are used to perform calculations. The two-body SU(3)-PDS Hamiltonian is as follows [57–60]:

Table 1. The RMS values for regular nuclei.

Nuclei	σ
$^{104}_{46}\text{Pd}$	0.194
$^{106}_{46}\text{Pd}$	0.112
$^{110}_{48}\text{Cd}$	0.124
$^{136}_{56}\text{Ba}$	0.081
$^{152}_{62}\text{Sm}$	0.142
$^{154}_{62}\text{Sm}$	0.131
$^{154}_{64}\text{Gd}$	0.152
$^{156}_{64}\text{Gd}$	0.161
$^{158}_{64}\text{Gd}$	0.184
$^{156}_{66}\text{Dy}$	0.125
$^{158}_{66}\text{Dy}$	0.147
$^{160}_{66}\text{Dy}$	0.176
$^{160}_{64}\text{Gd}$	0.118
$^{162}_{68}\text{Er}$	0.192
$^{164}_{68}\text{Er}$	0.178
$^{166}_{68}\text{Er}$	0.159
$^{168}_{70}\text{Yb}$	0.210
$^{168}_{68}\text{Er}$	0.118
$^{170}_{68}\text{Er}$	0.180
$^{170}_{70}\text{Yb}$	0.242
$^{172}_{70}\text{Yb}$	0.179
$^{174}_{72}\text{Hf}$	0.261
$^{176}_{72}\text{Hf}$	0.128
$^{178}_{72}\text{Hf}$	0.173
$^{180}_{72}\text{Hf}$	0.147
$^{182}_{74}\text{W}$	0.274
$^{184}_{74}\text{W}$	0.232
$^{186}_{74}\text{W}$	0.193
$^{186}_{76}\text{Os}$	0.129
$^{230}_{90}\text{Th}$	0.169
$^{232}_{90}\text{Th}$	0.116
$^{234}_{92}\text{U}$	0.078
$^{238}_{92}\text{U}$	0.093
$^{238}_{94}\text{Pu}$	0.076

$$\hat{H}_{PDS} = \hat{H}(h_0, h_2) + \hat{C}(O(3)) = h_0 P_0^\dagger P_0 + h_2 P_2^\dagger P_2 + \hat{C}(O(3)). \quad (3)$$

In equation (3), $P_0^\dagger = d^\dagger \cdot d^\dagger - 2(s^\dagger)^2$ (with $L=0$ angular momentum) and $P_2^\dagger = 2d^\dagger \cdot s^\dagger - \sqrt{7}(d^\dagger d^\dagger)$ (with $L=2$ angular momentum) represent the boson pair operator. Energy eigenvalues of SU(3)-PDS and other details are available in the [57–60].

To prove the proportionality of the dynamic limit and quantum number labeling, we report the root mean square (RMS) values in table 1. We consider the 0^+ , 2^+ , and 4^+ states from the first, second, and third bands for RMS calculation. RMS is a good criterion for the applicability of any model which is defined as [61]:

$$\sigma = \sqrt{\frac{\sum_{i=1}^n ((E_i)_{\text{exp}} - (E_i)_{\text{th}})^2}{(n-1)E(2_1^+)}}. \quad (4)$$

We label the eigenfunction of each nucleus with its corresponding dynamic limit and present the results for q_2 values in table 2. The reason for choosing 2_g^+ , 4_g^+ , 0_β^+ , 2_β^+ , 4_β^+ , 2_γ^+ , and 4_γ^+ levels is that we proved their regularity in [2]. Now, in this article, our goal is to study their regularity from the effective quadrupole deformation viewpoint.

The first four nuclei appearing in table 2 are not deformed. So, we use the U(5) dynamic limit for the calculation of the q_2 [22].

$$\begin{aligned}
q_{2U(5)} &= \langle n_{df}, n_{sf}, \tau_f, L_f | \hat{Q}^\chi \cdot \hat{Q}^\chi | n_{di}, n_{si}, \tau_i, L_i \rangle = \frac{1}{N_B} \\
&\times \langle J_n^+ | [(\hat{d}^\dagger \times \tilde{s} + \hat{s}^\dagger \times \tilde{d})^{(2)} + \chi(\hat{d}^\dagger \times \tilde{d})^{(2)}] \cdot [(\hat{d}^\dagger \times \tilde{s} + \hat{s}^\dagger \times \tilde{d})^{(2)} + \chi(\hat{d}^\dagger \times \tilde{d})^{(2)}] | J_n^+ \rangle \\
&= \frac{1}{N_B} \times \left[\langle 2.2; 0.2 | 2.2 \rangle \times (-1)^{n_d - L_i} \begin{pmatrix} L_i & 2 & L_f \\ -M_{L_i} & 0 & M_{L_f} \end{pmatrix} (-1)^{n_d - \tau_i} \begin{pmatrix} \tau_i & 2 & \tau_f \\ -M_{\tau_i} & 0 & M_{\tau_f} \end{pmatrix} (\sqrt{n_{di}} \sqrt{n_{si} + 1} + \sqrt{n_{si}} \sqrt{n_{di} + 1}) \right]^2,
\end{aligned} \quad (5)$$

Equation (5) is obtained by the effect of the creation and annihilation operators on the U(5) limit eigenfunction [11, 22, 23]. To calculate the q_2 , we use the Wigner–Eckart theorem and the Clebsch–Gordan coefficients [22]:

$$[T^{\kappa'} \times U^{\kappa''}]_q^\kappa = \sum_{q', q''} \langle \kappa' q' \kappa'' q'' | \kappa q \rangle T_q^{\kappa'} U_{q''}^{\kappa''}. \quad (6)$$

Also, N_B represents the number of bosons ($N_B = n_d + n_s$). To know how to determine quantum numbers τ , L and M , refer to [30–33].

To calculate the expectation value of q_2 in the SU(3) dynamic limit, we use the following equation [22]:

$$\begin{aligned}
\frac{q_2}{e^2_{SU(3)}} &= \langle n_d, n_s, K, (\lambda, \mu), L | \hat{Q}^\chi \cdot \hat{Q}^\chi | n_d, n_s, K, (\lambda, \mu), L \rangle \\
&= \left[\left[\langle 2.2; 0.2 | 2.2 \rangle \times (-1)^{n_{di} - L_i} \begin{pmatrix} L_i & 2 & L_f \\ -M_{L_i} & 0 & M_{L_f} \end{pmatrix} \times (-1)^{n_{di} - \tau_i} \begin{pmatrix} K_i & 2 & K_f \\ -M_{K_i} & 0 & M_{K_f} \end{pmatrix} \right. \right. \\
&\quad \times (-1)^{n_{di} - \lambda_i} \begin{pmatrix} \lambda_i & 2 & \lambda_f \\ -M_{\lambda_i} & 0 & M_{\lambda_f} \end{pmatrix} \times (\sqrt{n_{di}} \sqrt{n_{si} + 1} + \sqrt{n_{si}} \sqrt{n_{di} + 1}) \left. \right]^2 \\
&\quad + \left(2\chi \times \left[\left[\langle 2.2; 0.2 | 2.2 \rangle \times (-1)^{n_{di} - L_i} \begin{pmatrix} L_i & 2 & L_f \\ -M_{L_i} & 0 & M_{L_f} \end{pmatrix} \times (-1)^{n_{di} - K_i} \begin{pmatrix} K_i & 2 & K_f \\ -M_{K_i} & 0 & M_{K_f} \end{pmatrix} \right. \right. \\
&\quad \times (-1)^{n_{di} - \lambda_i} \begin{pmatrix} \lambda_i & 2 & \lambda_f \\ -M_{\lambda_i} & 0 & M_{\lambda_f} \end{pmatrix} \times (\sqrt{n_{di}} \sqrt{n_{di} + 1}) \cdot [\langle 2.2; 0.2 | 2.2 \rangle \\
&\quad \times (-1)^{n_{di} - L_i} \begin{pmatrix} L_i & 2 & L_f \\ -M_{L_i} & 0 & M_{L_f} \end{pmatrix} \times (-1)^{n_{di} - \tau_i} \begin{pmatrix} K_i & 2 & K_f \\ -M_{K_i} & 0 & M_{K_f} \end{pmatrix} \times (-1)^{n_{di} - \lambda_i} \begin{pmatrix} \lambda_i & 2 & \lambda_f \\ -M_{\lambda_i} & 0 & M_{\lambda_f} \end{pmatrix} \left. \right] \right] \\
&\quad \times (\sqrt{n_{di}} \sqrt{n_{si} + 1} + \sqrt{n_{si}} \sqrt{n_{di} + 1}) \left. \right]^2 \\
&\quad + \chi^2 [\langle 2.2; 0.2 | 2.2 \rangle \times (-1)^{n_{di} - L_i} \begin{pmatrix} L_i & 2 & L_f \\ -M_{L_i} & 0 & M_{L_f} \end{pmatrix} \\
&\quad \times (-1)^{n_{di} - K_i} \begin{pmatrix} K_i & 2 & K_f \\ -M_{K_i} & 0 & M_{K_f} \end{pmatrix} \times (-1)^{n_{di} - \lambda_i} \begin{pmatrix} \lambda_i & 2 & \lambda_f \\ -M_{\lambda_i} & 0 & M_{\lambda_f} \end{pmatrix} \\
&\quad \times (\sqrt{n_{di}} \sqrt{n_{di} + 1})^2]
\end{aligned} \quad (7)$$

Similarly with equations (5), (7) is obtained by the effect of the creation and annihilation operators on the SU(3) limit eigenfunction [11, 22, 23]. To know how to determine quantum numbers K , λ and L , refer to [30–33].

The relationship between q_2 and the effective quadrupole deformation is according to equation [56]

$$q_2(L_n^+) = [3ZeR_0^2/(4\pi)]^2 \beta_{eff}^2(L_n^+). \quad (8)$$

Where $R_0 = r_0 A^{\frac{1}{3}}$ ($r_0 = 1.2$ fm) indicates the nuclear radius and Z indicates the atomic number.

According to equation (8), we calculate the effective quadrupole deformation (β_{eff}) of each level and report it in table 3.

In tables 2 and 3, the sign ‘-’ means the absence of level experimentally.

The large quadrupole deformation of some levels in regular nuclei can be attributed to the interplay between the strong nuclear force and the electromagnetic force. In regular nuclei, the protons and neutrons are bound together by the strong nuclear force, which is a short-range force that acts to hold the nucleus together. However, the protons in the nucleus also repel each other due to the electromagnetic force, which is a long-range force that arises from the interaction between electric charges. The strong nuclear force tends to favor spherical shapes for nuclei, as this arrangement minimizes the repulsion between protons. However, the electromagnetic force can cause distortions in the nucleus, leading to non-spherical or deformed shapes. Specifically, quadrupole deformation refers to the deviation of the nucleus from a perfect sphere, where it becomes elongated along one axis and compressed along another. This deformation arises from the imbalance between the attractive strong nuclear force and the repulsive electromagnetic force. The large quadrupole deformation of some levels in

Table 2. The quadrupole shape constants value for different levels of regular nuclei.

Nuclei	$\frac{q_2}{e^2}(2_g^+)$	$\frac{q_2}{e^2}(4_g^+)$	$\frac{q_2}{e^2}(0_\beta^+)$	$\frac{q_2}{e^2}(2_\beta^+)$	$\frac{q_2}{e^2}(4_\beta^+)$	$\frac{q_2}{e^2}(2_\gamma^+)$	$\frac{q_2}{e^2}(4_\gamma^+)$
¹⁰⁴ ₄₆ Pd	0.017136	0.001926	0.71682	0.026118	0.001926	0.016788	0.003306
¹⁰⁶ ₄₆ Pd	0.020314	0.004669	0.878829	0.030317	0.004669	0.021679	0.00175
¹¹⁰ ₄₈ Cd	0.020314	0.081767	0.512645	0.050218	0.00175	0.05782	0.002723
¹³⁶ ₅₆ Ba	0.010772	0.000868	0.391804	0.01306	0.001492	0.006924	0.000113
¹⁵² ₆₂ Sm	0.00213	0.000158	0.04121	0.00155	0.0000528	0.0000118	0.0000075
¹⁵⁴ ₆₂ Sm	0.002288	0.00018	0.018513	0.000286	0.0000484	0.00000022	0.00001188
¹⁵⁴ ₆₄ Gd	0.002288	0.00018	0.046585	0.001958	0.0000231	0.00006039	0.00000275
¹⁵⁶ ₆₄ Gd	0.002424	0.000202	0.010776	0.0000648	0.00009576	0.00000372	0.0000084
¹⁵⁸ ₆₄ Gd	0.002548	0.000221	0.004927	0.0009945	0.00003497	0.000000013	0.000008073
¹⁵⁶ ₆₆ Dy	0.002424	0.000202	0.041112	0.001464	0.00000528	0.00003516	0.000000612
¹⁵⁸ ₆₆ Dy	0.002548	0.000221	0.032981	0.002691	0.000000026	0.00001599	0.00000884
¹⁶⁰ ₆₆ Dy	0.00266	0.000238	0.039928	0.0005866	0.000007	0.00005068	0.00002898
¹⁶⁰ ₆₄ Gd	0.00266	0.000238	0.015526	0.0013944	0.00004732	0.0000049	0.00000028
¹⁶² ₆₈ Er	0.002548	0.000221	0.017394	0.001872	0.000000026	0.0000247	0.00000884
¹⁶⁴ ₆₈ Er	0.00266	0.000238	0.001309	0.002268	0.00004732	0.0000049	0.00002898
¹⁶⁶ ₆₈ Er	0.00249	0.000224	0.001416	0.002115	0.0000441	0.0000471	0.000027
¹⁶⁸ ₇₀ Yb	0.00266	0.000238	0.010262	0.0013944	0.00000602	0.00005068	0.00001288
¹⁶⁸ ₆₈ Er	0.002864	0.000269	0.009888	0.003008	0.00010704	0.00000304	0.00000304
¹⁷⁰ ₆₈ Er	0.002941	0.000282	0.058837	0.001717	0.00002244	0.00005151	0.00000119
¹⁷⁰ ₇₀ Yb	0.00276	0.000237	0.031485	0.000942	0.00002565	0.0000159	0.00000675
¹⁷² ₇₀ Yb	0.002864	0.000269	0.03848	0.0013248	0.0000456	0.00003888	0.00000032
¹⁷⁴ ₇₂ Hf	0.00276	0.000254	0.04656	0.001785	0.0000225	0.000000015	0.00000675
¹⁷⁶ ₇₂ Hf	0.002864	0.000269	0.03848	0.0013248	0.00005504	0.000003184	0.00002256
¹⁷⁸ ₇₂ Hf	0.00276	0.000237	0.0165	0.001785	0.0000000225	0.0000159	0.00000675
¹⁸⁰ ₇₂ Hf	0.00266	0.000238	0.024402	0.0005866	0.0002254	0.00000497	0.00002898
¹⁸² ₇₄ W	0.002548	0.000221	0.032981	0.0009945	0.000000026	0.00001599	0.00000195
¹⁸⁴ ₇₄ W	0.002424	0.000202	0.025788	0.00234	0.000005328	0.00001296	0.00000048
¹⁸⁶ ₇₄ W	0.002288	0.00018	0.018513	0.001958	0.0000231	0.00001628	0.000000011
¹⁸⁶ ₇₆ Os	0.002288	0.00018	0.018513	0.002662	0.00008547	0.00001628	0.000000033
²³⁰ ₉₀ Th	0.002288	0.00018	0.018513	0.000286	0.000000352	0.00003454	0.000000011
²³² ₉₀ Th	0.002424	0.000202	0.010778	0.0006156	0.000010848	0.000003768	0.000000612
²³⁴ ₉₂ U	0.002548	0.000221	0.017394	0.0001976	0.000000035	0.00001599	0.000002002
²³⁸ ₉₂ U	0.00276	0.000254	0.031485	0.000942	0.00002565	0.000001575	0.00000066
²³⁸ ₉₄ Pu	0.00276	0.000254	0.031485	0.000942	0.0000000225	0.0000159	—

regular nuclei occurs when the repulsion between protons becomes significant enough to overcome the stabilizing effect of the strong nuclear force. This can happen when there is an excess of protons or when the number of neutrons is not sufficient to counterbalance the electromagnetic repulsion. Furthermore, certain energy levels in nuclei can have specific shapes due to quantum mechanical effects. These shapes can be influenced by the interplay between the nuclear forces and the electromagnetic forces, leading to large quadrupole deformations at some levels.

In this article, our goal is to study the reason for the existence of regularity in regular nuclei. According to previous studies [1, 2], the origin of regularity depends on the type of nuclear structure. On this basis, Alhassid and Whelan introduced the region of the arc of regularity. In this regard, we schematically show the dependence of our desired observable (β_{eff}) on the control parameters of the Hamiltonian in a selected nucleus.

We choose 2_g^+ , 2_β^+ , and 2_γ^+ levels, and show (in figure 1) the β_{eff} changes in terms of η and χ for the ¹⁶⁴Er nucleus (with $N_B = 14$).

As we can see in figure 1, the amount of β_{eff} (in 2_g^+ , 2_β^+ , and 2_γ^+ levels) changes with the change of Hamiltonian control parameters. So we conclude that the amount of β_{eff} for nuclei belonging to the region of the arc of regularity (in the IBM symmetry triangle) is different from other regions.

Now that the values of β_{eff} have been determined for each level, we define the patterns of equation (9) to study the regularity:

$$R_{0,2}^{\beta,g} = \frac{|\beta_{eff}(0_\beta^+) - \beta_{eff}(2_g^+)|}{\beta_{eff}(2_g^+)} \quad (9)$$

Table 3. The effective electric quadruple deformation values for different levels of regular nuclei (all in b).

Nuclei	$\beta_{eff}(2_g^+)$	$\beta_{eff}(4_g^+)$	$\beta_{eff}(0_\beta^+)$	$\beta_{eff}(2_\beta^+)$	$\beta_{eff}(4_\beta^+)$	$\beta_{eff}(2_\gamma^+)$	$\beta_{eff}(4_\gamma^+)$
¹⁰⁴ ₄₆ Pd	0.827375	0.2773806	5.3512238	1.0214521	0.277380676	0.818931433	0.363412186
¹⁰⁶ ₄₆ Pd	0.900836	0.4318769	5.9251622	1.1005028	0.431876922	0.930609933	0.26440343
¹¹⁰ ₄₈ Cd	0.863301	1.7320231	4.3368361	1.3573578	0.253386621	1.456477228	0.316073784
¹³⁶ ₅₆ Ba	0.538847	0.1529599	3.2497675	0.5933206	0.200540585	0.432012582	0.05518958
¹⁵² ₆₂ Sm	0.216423	0.0589444	0.9519530	0.1846205	0.034074631	0.016108504	0.01284236
¹⁵⁴ ₆₂ Sm	0.224306	0.0629144	0.6380469	0.0793043	0.032623976	0.002199508	0.016163017
¹⁵⁴ ₆₄ Gd	0.217297	0.0609483	0.9805030	0.2010167	0.02183393	0.035302742	0.007533421
¹⁵⁶ ₆₄ Gd	0.223662	0.0645656	0.4715792	0.0365690	0.044454731	0.008761882	0.013166355
¹⁵⁸ ₆₄ Gd	0.229311	0.0675339	0.3188725	0.1432611	0.026864189	0.000517962	0.012907537
¹⁵⁶ ₆₆ Dy	0.216884	0.0626091	0.8931949	0.1685514	0.010122293	0.026120795	0.003446178
¹⁵⁸ ₆₆ Dy	0.222362	0.0654874	0.8000071	0.2285171	0.000710311	0.017615144	0.01309749
¹⁶⁰ ₆₆ Dy	0.227197	0.0679595	0.8802392	0.1066922	0.011654965	0.031360304	0.023714346
¹⁶⁰ ₆₄ Gd	0.234297	0.0700832	0.5660511	0.1696366	0.031249875	0.01005597	0.002403837
¹⁶² ₆₈ Er	0.215822	0.0635613	0.5638930	0.1849907	0.00068942	0.021249341	0.01271227
¹⁶⁴ ₆₈ Er	0.220514	0.0659607	0.1546916	0.2036190	0.029411647	0.009464442	0.023016865
¹⁶⁶ ₆₈ Er	0.213351	0.0639913	0.1608898	0.1966310	0.028393326	0.029343197	0.022216665
¹⁶⁸ ₇₀ Yb	0.214214	0.0640761	0.4207497	0.1550963	0.010190749	0.029568286	0.01490616
¹⁶⁸ ₆₈ Er	0.228814	0.0701250	0.4251588	0.2344962	0.044235407	0.007454762	0.007454762
¹⁷⁰ ₆₈ Er	0.231869	0.0717995	1.0371038	0.1771667	0.020253888	0.030686175	0.004664128
¹⁷⁰ ₇₀ Yb	0.218203	0.0639413	0.7369867	0.1274774	0.021035437	0.016561757	0.010790951
¹⁷² ₇₀ Yb	0.218203	0.0639413	0.7369867	0.1274774	0.021035437	0.016561757	0.010790951
¹⁷⁴ ₇₂ Hf	0.212142	0.0643561	0.8713244	0.1706052	0.019154228	0.00049456	0.010491203
¹⁷⁶ ₇₂ Hf	0.216102	0.0662292	0.7921199	0.1469767	0.029957986	0.00720543	0.01917975
¹⁷⁸ ₇₂ Hf	0.212142	0.0621652	0.5186989	0.1706052	0.00060571	0.016101708	0.010491203
¹⁸⁰ ₇₂ Hf	0.208264	0.0622962	0.6307918	0.0978012	0.060624805	0.009002261	0.02173815
¹⁸² ₇₄ W	0.198323	0.0584077	0.7135198	0.1239015	0.000633521	0.015710804	0.005486451
¹⁸⁴ ₇₄ W	0.193437	0.0558405	0.6309326	0.1900562	0.009068935	0.014144144	0.002722042
¹⁸⁶ ₇₄ W	0.187932	0.0527121	0.5345798	0.1738523	0.018883399	0.015852632	0.00041207
¹⁸⁶ ₇₆ Os	0.182987	0.0513249	0.5205119	0.1973769	0.035367076	0.015435457	0.000694943
²³⁰ ₉₀ Th	0.154522	0.0433410	0.4395434	0.0546319	0.001916614	0.01898561	0.000338813
²³² ₉₀ Th	0.159048	0.0459133	0.3353763	0.0801516	0.010639917	0.006270741	0.002527197
²³⁴ ₉₂ U	0.159520	0.0469801	0.4167904	0.0444233	0.000591224	0.012636951	0.004471468
²³⁸ ₉₂ U	0.166024	0.0503657	0.5607508	0.0969936	0.016005224	0.003966051	0.002567379
²³⁸ ₉₄ Pu	0.162492	0.0492940	0.5488199	0.0949300	0.000463948	0.012333223	—

$$R_{2,2}^{\beta,g} = \frac{|\beta_{eff}(2_\beta^+) - \beta_{eff}(2_g^+)|}{\beta_{eff}(2_g^+)}$$

$$R_{4,2}^{\beta,g} = \frac{|\beta_{eff}(4_\beta^+) - \beta_{eff}(2_g^+)|}{\beta_{eff}(2_g^+)}$$

$$R_{0,4}^{\beta,g} = \frac{|\beta_{eff}(0_\beta^+) - \beta_{eff}(4_g^+)|}{\beta_{eff}(4_g^+)}$$

$$R_{2,4}^{\beta,g} = \frac{|\beta_{eff}(2_\beta^+) - \beta_{eff}(4_g^+)|}{\beta_{eff}(4_g^+)}$$

$$R_{4,4}^{\beta,g} = \frac{|\beta_{eff}(4_\beta^+) - \beta_{eff}(4_g^+)|}{\beta_{eff}(4_g^+)}$$

$$R_{2,2}^{\gamma,g} = \frac{|\beta_{eff}(2_\gamma^+) - \beta_{eff}(2_g^+)|}{\beta_{eff}(2_g^+)}$$

$$R_{4,2}^{\gamma,g} = \frac{|\beta_{eff}(4_\gamma^+) - \beta_{eff}(2_g^+)|}{\beta_{eff}(2_g^+)}$$

$$R_{2,4}^{\gamma,g} = \frac{|\beta_{eff}(2_\gamma^+) - \beta_{eff}(4_g^+)|}{\beta_{eff}(4_g^+)}$$

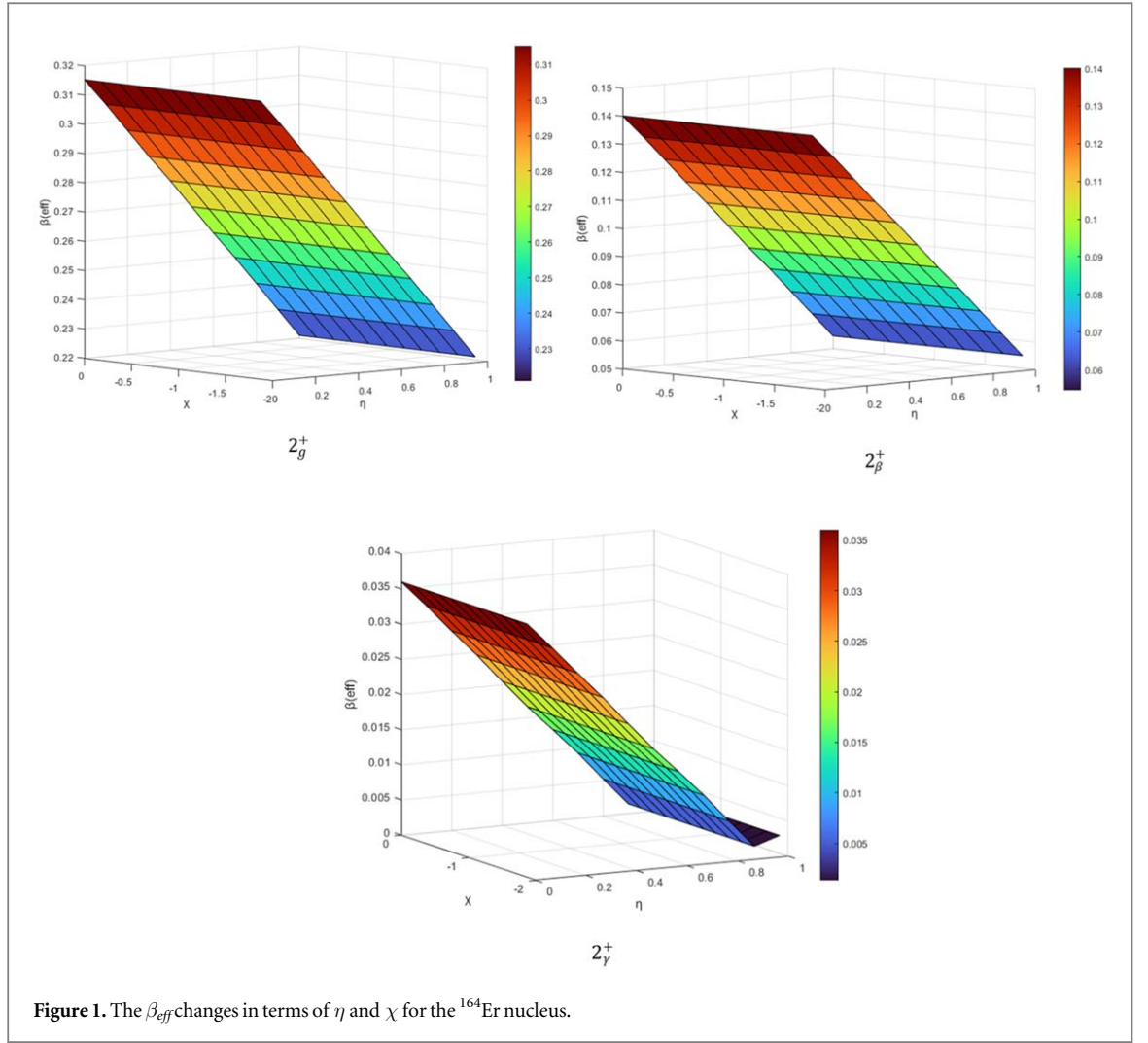


Figure 1. The β_{eff} changes in terms of η and χ for the ^{164}Er nucleus.

$$\begin{aligned}
 R_{4,4}^{\gamma,g} &= \frac{|\beta_{eff}(4_\gamma^+) - \beta_{eff}(4_g^+)|}{\beta_{eff}(4_g^+)} \\
 R_{2,0}^{\gamma,\beta} &= \frac{|\beta_{eff}(2_\gamma^+) - \beta_{eff}(0_\beta^+)|}{\beta_{eff}(0_\beta^+)} \\
 R_{2,2}^{\gamma,\beta} &= \frac{|\beta_{eff}(2_\gamma^+) - \beta_{eff}(2_\beta^+)|}{\beta_{eff}(2_\beta^+)} \\
 R_{2,4}^{\gamma,\beta} &= \frac{|\beta_{eff}(2_\gamma^+) - \beta_{eff}(4_\beta^+)|}{\beta_{eff}(4_\beta^+)} \\
 R_{4,0}^{\gamma,\beta} &= \frac{|\beta_{eff}(4_\gamma^+) - \beta_{eff}(0_\beta^+)|}{\beta_{eff}(0_\beta^+)} \\
 R_{4,2}^{\gamma,\beta} &= \frac{|\beta_{eff}(4_\gamma^+) - \beta_{eff}(2_\beta^+)|}{\beta_{eff}(2_\beta^+)} \\
 R_{4,4}^{\gamma,\beta} &= \frac{|\beta_{eff}(4_\gamma^+) - \beta_{eff}(4_\beta^+)|}{\beta_{eff}(4_\beta^+)}
 \end{aligned}$$

We present the values related to patterns in table 4.

The results of table 4 indicate that the defined ratios are within a certain range (Especially evident for the ratios with higher spin and band levels). The reason for this is that more transitions occur in the lower spin levels in the ground band. Therefore, the levels with higher spin have fewer changes, and the repetition pattern is more visible in them. In [2], we related the existence of repetition patterns (for the quadrupole transition probabilities and energy levels) to the similarity of the structures of the corresponding nuclei. Therefore, in this article, we

Table 4. The ratios of the effective quadruple deformation values for different levels of regular nuclei.

Nuclei	$R_{0,2}^{\beta,g}$	$R_{2,2}^{\beta,g}$	$R_{4,2}^{\beta,g}$	$R_{0,4}^{\beta,g}$	$R_{2,4}^{\beta,g}$	$R_{4,4}^{\beta,g}$	$R_{2,2}^{\gamma,g}$	$R_{4,2}^{\gamma,g}$	$R_{2,4}^{\gamma,g}$	$R_{4,4}^{\gamma,g}$	$R_{2,0}^{\gamma,\beta}$	$R_{2,2}^{\gamma,\beta}$	$R_{2,4}^{\gamma,\beta}$	$R_{4,0}^{\gamma,\beta}$	$R_{4,2}^{\gamma,\beta}$	$R_{4,4}^{\gamma,\beta}$
$^{104}_{46}\text{Pd}$	5.46	0.23	0.66	18.2	2.68	0.00000027	0.01	0.56	1.95	0.31	0.84	0.19	1.95	0.94	0.64	0.31
$^{106}_{46}\text{Pd}$	5.57	0.22	0.52	12.7	1.54	0.00000005	0.03	0.70	1.15	0.38	0.84	0.15	1.15	0.92	0.75	0.38
$^{110}_{48}\text{Cd}$	4.02	0.57	0.70	1.50	0.21	0.85	0.68	0.63	0.15	0.81	0.66	0.07	4.74	0.94	0.76	0.24
$^{136}_{56}\text{Ba}$	5.03	0.10	0.62	20.2	2.87	0.31	0.19	0.89	1.82	0.63	0.86	0.27	1.15	0.93	0.90	0.72
$^{152}_{62}\text{Sm}$	3.39	0.14	0.84	15.1	2.13	0.42	0.92	0.94	0.72	0.78	0.98	0.91	0.52	0.96	0.93	0.62
$^{154}_{62}\text{Sm}$	1.84	0.64	0.85	9.14	0.26	0.48	0.99	0.92	0.96	0.74	0.99	0.97	0.93	0.94	0.79	0.50
$^{154}_{64}\text{Gd}$	3.51	0.07	0.89	15.0	2.29	0.64	0.83	0.96	0.42	0.87	0.96	0.82	0.61	0.97	0.96	0.65
$^{156}_{64}\text{Gd}$	1.10	0.83	0.80	6.30	0.43	0.31	0.96	0.94	0.86	0.79	0.98	0.76	0.80	0.90	0.63	0.70
$^{158}_{64}\text{Gd}$	0.39	0.37	0.88	3.72	1.12	0.60	0.99	0.94	0.99	0.80	0.99	0.99	0.98	0.91	0.90	0.51
$^{156}_{66}\text{Dy}$	3.11	0.22	0.95	13.2	1.69	0.83	0.87	0.98	0.58	0.94	0.97	0.84	1.58	0.98	0.97	0.65
$^{158}_{66}\text{Dy}$	2.59	0.02	0.99	11.2	2.48	0.98	0.92	0.94	0.73	0.79	0.97	0.92	23.7	0.99	0.94	17.43
$^{160}_{66}\text{Dy}$	2.87	0.53	0.94	11.9	0.56	0.82	0.86	0.89	0.53	0.65	0.96	0.70	1.69	0.98	0.77	1.03
$^{160}_{64}\text{Gd}$	1.41	0.27	0.86	7.07	1.42	0.55	0.95	0.98	0.85	0.96	0.98	0.94	0.67	0.94	0.98	0.92
$^{162}_{68}\text{Er}$	1.61	0.14	0.99	7.87	1.91	0.98	0.90	0.94	0.66	0.79	0.96	0.88	29.8	0.99	0.93	17.43
$^{164}_{68}\text{Er}$	0.29	0.07	0.86	1.34	2.08	0.55	0.95	0.89	0.85	0.65	0.93	0.95	0.67	0.80	0.88	0.21
$^{166}_{68}\text{Er}$	0.24	0.07	0.86	1.51	2.07	0.55	0.86	0.89	0.54	0.65	0.81	0.85	0.03	0.82	0.88	0.21
$^{168}_{70}\text{Yb}$	0.96	0.27	0.95	5.56	1.42	0.84	0.86	0.93	0.53	0.76	0.92	0.80	1.90	0.97	0.90	0.46
$^{168}_{68}\text{Er}$	0.85	0.02	0.80	5.06	2.34	0.36	0.96	0.96	0.89	0.89	0.98	0.96	0.83	0.89	0.96	0.83
$^{170}_{68}\text{Er}$	3.47	0.23	0.91	13.4	1.46	0.71	0.86	0.97	0.57	0.93	0.97	0.82	0.51	0.98	0.97	0.76
$^{170}_{70}\text{Yb}$	2.37	0.41	0.90	10.5	0.99	0.67	0.92	0.95	0.74	0.83	0.97	0.87	0.21	0.97	0.91	0.48
$^{172}_{70}\text{Yb}$	2.37	0.41	0.90	10.5	0.99	0.67	0.92	0.95	0.74	0.83	0.97	0.87	0.21	0.97	0.91	0.48
$^{174}_{72}\text{Hf}$	3.10	0.19	0.90	12.5	1.65	0.70	0.99	0.95	0.99	0.83	0.99	0.99	0.97	0.97	0.93	0.45
$^{176}_{72}\text{Hf}$	2.66	0.31	0.86	10.9	1.21	0.54	0.96	0.91	0.89	0.71	0.99	0.95	0.75	0.96	0.86	0.35
$^{178}_{72}\text{Hf}$	1.44	0.19	0.99	7.34	1.74	0.99	0.92	0.95	0.74	0.83	0.96	0.90	25.5	0.99	0.93	16.32
$^{180}_{72}\text{Hf}$	2.02	0.53	0.70	9.12	0.56	0.02	0.95	0.89	0.85	0.65	0.98	0.90	0.85	0.90	0.77	0.64
$^{182}_{74}\text{W}$	2.59	0.37	0.99	11.2	1.12	0.98	0.92	0.97	0.73	0.90	0.97	0.87	23.7	0.99	0.95	7.66
$^{184}_{74}\text{W}$	2.26	0.01	0.95	10.2	2.40	0.83	0.92	0.98	0.74	0.95	0.97	0.92	0.55	0.98	0.98	0.69
$^{186}_{74}\text{W}$	1.84	0.07	0.89	9.14	2.29	0.64	0.91	0.99	0.69	0.99	0.97	0.90	0.16	0.96	0.99	0.97
$^{186}_{76}\text{Os}$	1.84	0.07	0.80	9.14	2.84	0.31	0.91	0.99	0.69	0.98	0.97	0.92	0.56	0.93	0.99	0.98
$^{230}_{90}\text{Th}$	1.84	0.64	0.98	9.14	0.26	0.95	0.87	0.99	0.56	0.99	0.95	0.65	8.90	0.99	0.99	0.82
$^{232}_{90}\text{Th}$	1.10	0.49	0.93	6.30	0.74	0.76	0.96	0.98	0.86	0.94	0.98	0.92	0.41	0.96	0.96	0.76
$^{234}_{92}\text{U}$	1.61	0.72	0.99	7.87	0.05	0.98	0.92	0.97	0.73	0.90	0.96	0.71	20.3	0.99	0.89	6.56
$^{238}_{92}\text{U}$	2.37	0.41	0.90	10.1	0.92	0.68	0.97	0.98	0.92	0.94	0.99	0.95	0.75	0.97	0.97	0.83
$^{238}_{94}\text{Pu}$	2.37	0.41	0.99	10.1	0.92	0.99	0.92	—	0.74	—	0.97	0.87	25.5	0.99	—	—

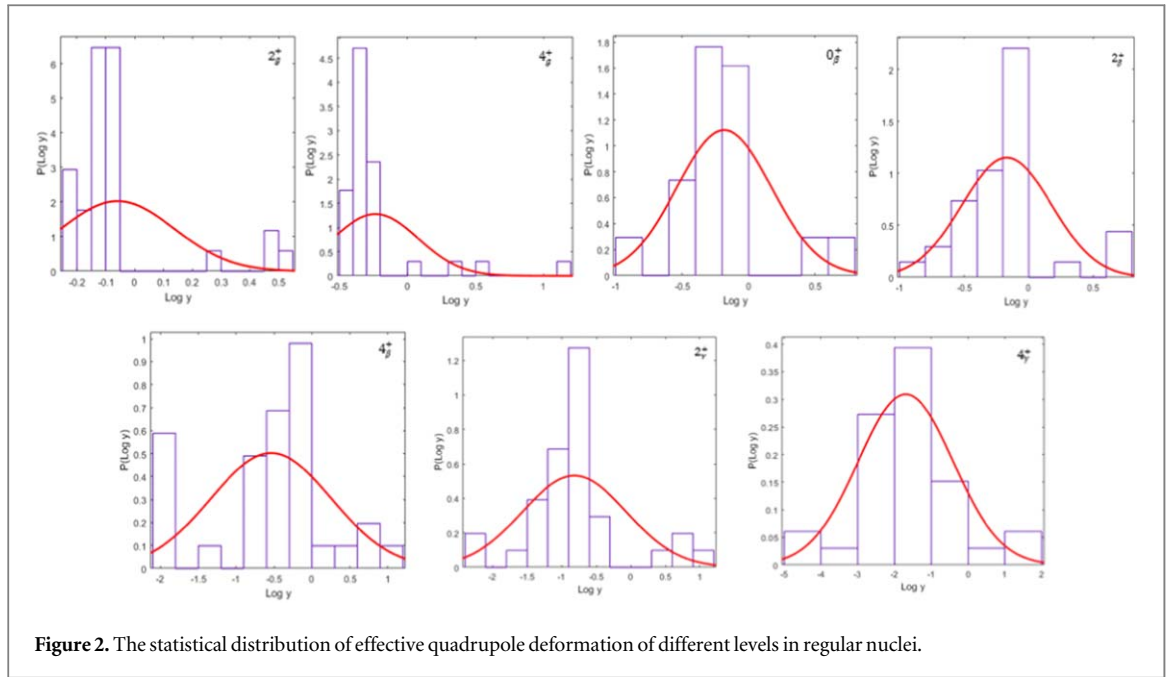


Figure 2. The statistical distribution of effective quadrupole deformation of different levels in regular nuclei.

Table 5. The values of the q parameter for effective quadrupole deformation of different levels of the regular nuclei.

β_{eff}	q
$\beta_{eff}(2_g^+)$	0.1286
$\beta_{eff}(4_g^+)$	0.4351
$\beta_{eff}(0_\beta^+)$	0.5312
$\beta_{eff}(2_\beta^+)$	0.5307
$\beta_{eff}(4_\beta^+)$	0.6836
$\beta_{eff}(2_\gamma^+)$	0.7359
$\beta_{eff}(4_\gamma^+)$	0.6327

conclude that the reason for the regularity of the corresponding nuclei is the similarity of the effective quadrupole deformation of the levels involved in electromagnetic transitions.

The repetition patterns observed in regular nuclei have significant statistical significance. These patterns can be understood through the principles of quantum mechanics and the underlying symmetries of the nuclear forces. The statistical significance of these repetition patterns lies in their consistency and predictability. Experimental data on energy levels and other properties of regular nuclei consistently match the predictions of theoretical models. This agreement confirms the underlying symmetries and regularities in nuclear structure. Moreover, the statistical behavior of energy levels in regular nuclei can be analyzed using random matrix theory (RMT). RMT provides a theoretical framework for understanding the statistical properties of energy spectra in systems with chaotic or random behavior. The success of RMT in explaining various statistical properties of energy levels in regular nuclei further highlights the statistical significance of repetition patterns.

For further study, we check the regularity of our desired nuclei (from the perspective of β_{eff}) using RMT. For this purpose, we first unfold the data related to each level. The method used to unfold the data for the statistical study of energy levels is the nearest neighbor spacing distribution (NNSD) [62–65]. However, regarding β_{eff} , the issue of the distance between the data is not discussed. For this reason, we use the method used in [8, 66] for unfolding.

To analyze the data, RMT was applied. RMT is a mathematical framework that allows for the analysis of complex systems with random interactions. In the context of nuclear physics, RMT has been successfully used to study the statistical properties of energy levels and other observables in atomic nuclei. In this study, RMT was

used to analyze the deformation of the levels and their fluctuations in the nucleus under investigation. The deformation of the levels was obtained through theoretical calculations. The main idea behind RMT is that the statistical properties of these levels can be described by random matrices that have certain symmetry properties. The first step in applying RMT is to construct a suitable ensemble of random matrices that captures the relevant symmetries of the system. In the case of atomic nuclei, the appropriate ensemble is typically the Gaussian Orthogonal Ensemble (GOE) or the Gaussian Unitary Ensemble (GUE), depending on the presence or absence of time-reversal symmetry, respectively. Once the ensemble of random matrices is constructed, various statistical measures can be calculated and compared with the experimental or theoretical data. These measures include level spacing distributions, nearest-neighbor spacing distributions, and spectral rigidity, among others. By comparing these measures, one can determine whether the levels follow the expected statistical patterns predicted by RMT.

To determine whether the system is regular or disordered (by determining the degree of similarity with Gaussian ($P(s) = \frac{\pi s}{2} \exp(-\frac{\pi s^2}{4})$) and Poisson ($P(s) = e^{-s}$) distributions), we use the fitting of the Berry-Robnik distribution (BRD) [67–69] distribution function to the data. The BRD probability distribution function is according to the following equation:

$$P(y) = \left[q + \frac{\pi}{2}(1 - q)y \right] \exp\left(-qy - \frac{\pi}{4}(1 - q)y^2\right). \quad (10)$$

In equation (10), the parameter $q = 0$ represents the Gaussian distribution, and $q = 1$ represents the Poisson distribution. We have shown the statistical distributions of the β_{eff} for different levels in figure 2.

Also, we presented q values (which we obtained from the curve fitting method in MATLAB software) in table 5.

The results of the statistical study in figure 2 (as well as the q values in table 5) indicate that there is less regularity for the levels of the ground band compared to the levels belonging to the beta and gamma bands. That is why the ratio of beta to gamma bands is more regular. In other words, the results obtained from the study of different ratios (equation (10)) were also confirmed by studying the statistical distribution of the effective quadrupole deformations.

We also study the relationship between regularity and triaxiality. Besides quadruple deformation (β_2), it is also possible to describe the shape of the nuclei using the γ parameter [70]. According to the following relationship, the γ angle is obtained from the $\frac{E(2_1^+)}{E(2_1^+)}$ ratio [70–73].

$$\sin(3\gamma) = \frac{3}{2\sqrt{2}} \sqrt{1 - \left(\frac{R-1}{R+1}\right)^2}; R = \frac{E(2_2^+)}{E(2_1^+)}. \quad (11)$$

We obtain R values from empirical sources [74]. We report the γ values in table 6.

The energy levels we used in the study are according to table 7.

The gamma values reported in table 6 are particularly timely, in view of recent attention paid to triaxiality appearing in shell model calculations as a consequence of the tensor force [75, 76], as well as in Triaxial Projected Shell Model (TPSM) calculations [77].

The value of $\beta_2 = 0$ represents spherical nuclei. If $\beta_2 > 0$, then for different values of γ [70]:

$$\begin{cases} \gamma = 0^\circ & \rightarrow \text{Prolate nuclei} \\ \gamma = 60^\circ & \rightarrow \text{Oblate nuclei} \\ 0^\circ < \gamma < 60^\circ & \rightarrow \text{Triaxial nucle} \end{cases}$$

Also, $\gamma = 30^\circ$ represents rigid triaxial nuclei. Therefore, according to the γ values in table 6, we conclude that the nuclei under our investigation are in the triaxial range (closer to the prolate limit). In addition to the repetition patterns, regular nuclei also exhibit triaxial properties. This refers to the deformation of the nucleus into a non-spherical shape, where the axes of rotation are not equal. The triaxial properties of regular nuclei can be understood through the principles of quantum mechanics and the interplay between nuclear forces and the motion of nucleons. The underlying symmetries of the nuclear forces allow for different shapes and orientations of the nucleus, leading to triaxial deformation. Experimental evidence for triaxiality in regular nuclei comes from measurements of various observables, such as electromagnetic transitions and nuclear moments. These measurements reveal the presence of non-zero electric quadrupole moments, which indicate a departure from spherical symmetry. The statistical significance of triaxial properties lies in their prevalence and consistency across different nuclei. Triaxial deformation has been observed in a wide range of nuclei, from light to heavy isotopes. This indicates that it is a common feature of nuclear structure.

The repetition pattern of energy levels in regular nuclei is significant because it provides insights into the underlying nuclear structure and regularity. The observed pattern suggests that there are certain symmetries and interactions within the nucleus that lead to the repetition of energy levels. One potential physical reason behind

Table 6. The γ values for regular nuclei (in deg).

Nuclei	R	γ
$^{104}_{46}\text{Pd}$	2.4139	24.9626
$^{106}_{46}\text{Pd}$	2.2038	26.4670
$^{110}_{48}\text{Cd}$	2.2436	26.1362
$^{136}_{56}\text{Ba}$	1.8948	32.4485
$^{152}_{62}\text{Sm}$	6.6549	15.2113
$^{154}_{62}\text{Sm}$	14.3668	10.5166
$^{154}_{64}\text{Gd}$	6.6261	15.2426
$^{156}_{64}\text{Gd}$	12.6945	11.1660
$^{158}_{64}\text{Gd}$	14.9299	10.3222
$^{156}_{66}\text{Dy}$	6.0146	15.9577
$^{158}_{66}\text{Dy}$	9.5667	12.7948
$^{160}_{66}\text{Dy}$	11.1325	11.8962
$^{160}_{64}\text{Gd}$	13.1345	10.9835
$^{162}_{68}\text{Er}$	8.8271	13.2973
$^{164}_{68}\text{Er}$	9.4139	12.8939
$^{166}_{68}\text{Er}$	9.7533	12.6769
$^{168}_{70}\text{Yb}$	11.2162	11.8534
$^{168}_{68}\text{Er}$	10.2898	12.3552
$^{170}_{68}\text{Er}$	11.8847	11.5271
$^{170}_{70}\text{Yb}$	13.5131	10.8335
$^{172}_{70}\text{Yb}$	14.1965	8.1339
$^{174}_{72}\text{Hf}$	9.8943	12.5899
$^{176}_{72}\text{Hf}$	13.8839	10.6924
$^{178}_{72}\text{Hf}$	12.6059	11.2039
$^{180}_{72}\text{Hf}$	12.6801	11.1721
$^{182}_{74}\text{W}$	12.2010	11.3819
$^{184}_{74}\text{W}$	8.1219	13.8366
$^{186}_{74}\text{W}$	6.0177	15.9538
$^{186}_{76}\text{Os}$	5.5955	16.5119
$^{230}_{90}\text{Th}$	12.7287	11.1515
$^{232}_{90}\text{Th}$	15.6808	10.0791
$^{234}_{92}\text{U}$	19.5810	9.0451
$^{238}_{92}\text{U}$	21.5097	8.6390
$^{238}_{94}\text{Pu}$	22.3099	8.4859

this repetition pattern is the presence of collective motion in regular nuclei. Collective motion refers to the coordinated motion of a large number of nucleons within the nucleus. This motion can give rise to specific patterns in energy levels due to the collective behavior of nucleons. Another possible reason for the repetition pattern is the interplay between collective and single-particle motion. While collective models can explain some aspects of nuclear behavior, they do not fully account for the role of individual nucleons and their interactions. The repetition pattern may arise from the combination of collective motion and the specific arrangement of nucleons in the nucleus. Furthermore, the distribution of protons and neutrons within the nucleus, which is responsible for quadrupole deformation, can also contribute to the repetition pattern. The specific arrangement and interactions of nucleons can lead to certain energy levels being repeated across different nuclei with quadrupole deformation. Understanding the physical reasons behind the repetition pattern of energy levels in regular nuclei is crucial for developing a comprehensive understanding of nuclear structure and regularity. It allows us to uncover the underlying symmetries and interactions within the nucleus, which in turn can provide insights into other nuclear properties and phenomena.

In our previous work [78], in which the spectral statistics of even–even nuclei are analyzed using the energy spectra, a significant relation is observed between the degrees of deformation and statistical fluctuation of these nuclei. In the current study, the quadrupole deformation is used for statistical analyses, and the triaxiality of the considered ‘regular nuclei’ is determined, and the reported pattern confirms the possible correlation between these quantities, as suggested by our previous results. For results presented in this study, there are no other counterparts because this is the first attempt at such consideration from this viewpoint [79] is the only possible source for comparing deformation values, which does not include all the levels we have examined and is only for the ground band. Therefore, it is not possible to compare their nature.’

Table 7. The 2_g^+ , 4_g^+ , 0_β^+ , 2_β^+ , 4_β^+ , 2_γ^+ , and 4_γ^+ energy levels for regular nuclei (in keV).

Nuclei	$E(2_g^+)$	$E(4_g^+)$	$E(2_\gamma^+)$	$E(4_\gamma^+)$	$E(0_\beta^+)$	$E(2_\beta^+)$	$E(4_\beta^+)$
$^{104}_{46}\text{Pd}$	555.81	1323.59	1341.68	2082.38	1792.86	1794.3	2181.56
$^{106}_{46}\text{Pd}$	511.85	1229.30	1128.02	1932.32	1706.44	1562.25	2076.69
$^{110}_{48}\text{Cd}$	657.76	1542.44	1475.79	1809.48	1731.31	1783.49	2220.06
$^{136}_{56}\text{Ba}$	818.52	1866.61	1550.98	2053.89	2141.38	2080.13	2356.49
$^{152}_{62}\text{Sm}$	121.78	366.47	810.45	1022.97	1082.84	1085.84	1371.73
$^{154}_{62}\text{Sm}$	81.98	266.81	1177.81	1337.6	1202.44	1286.29	1472.16
$^{154}_{64}\text{Gd}$	123.07	370.99	815.49	1047.59	1182.09	996.25	1263.77
$^{156}_{64}\text{Gd}$	88.97	288.18	1129.43	1297.82	1168.18	1154.15	1355.42
$^{158}_{64}\text{Gd}$	79.51	261.45	1187.14	1358.47	1452.35	1259.87	1380.63
$^{156}_{66}\text{Dy}$	137.77	404.19	828.64	1088.28	1377.80	890.50	1168.47
$^{158}_{66}\text{Dy}$	98.91	317.13	946.32	1163.75	1269	1085.55	1280.01
$^{160}_{66}\text{Dy}$	86.78	283.82	966.16	1155.84	703	1349.75	1522.33
$^{160}_{64}\text{Gd}$	75.26	248.50	988.54	1070.42	1558.35	1325.7	1147.98
$^{162}_{68}\text{Er}$	102.04	329.62	900.72	1128.11	2114.11	1171.02	1369
$^{164}_{68}\text{Er}$	91.38	299.43	860.25	1058.49	1416.57	1314.56	1469.72
$^{166}_{68}\text{Er}$	80.57	264.99	785.90	956.23	1713.4	1528.40	1678.76
$^{168}_{70}\text{Yb}$	87.73	286.55	984	1171.38	1197	1233.1	1390.12
$^{168}_{68}\text{Er}$	79.80	264.08	821.16	994.74	1422.12	1276.27	1411.09
$^{170}_{68}\text{Er}$	78.59	260.14	934.02	1103.36	1324.26	959.99	1127.29
$^{170}_{70}\text{Yb}$	84.25	277.43	1138.55	1292.4	1228.84	1145.72	1329.31
$^{172}_{70}\text{Yb}$	78.74	260.26	1117.87	1263.02	1405.008	1465.87	1286.54
$^{174}_{72}\text{Hf}$	90.98	297.38	900.24	1062.17	—	1226.77	1503.29
$^{176}_{72}\text{Hf}$	88.34	290.18	1226.63	1390.19	1293.12	1341.31	1540.3
$^{178}_{72}\text{Hf}$	93.18	306.61	1174.63	1384.46	1434.23	1276.69	1450.36
$^{180}_{72}\text{Hf}$	93.32	308.57	1183.36	1369.65	1315.7	1199.72	1381.58
$^{182}_{74}\text{W}$	100.10	329.42	1221.40	1442.83	1265	1257.41	1510.22
$^{184}_{74}\text{W}$	111.21	364.06	903.30	1133.85	1322.15	1121.44	1360.38
$^{186}_{74}\text{W}$	122.63	396.54	737.96	1006.59	1150	1014.97	1006.73
$^{186}_{76}\text{Os}$	137.15	434.08	767.47	1070.48	1456	1208.44	1351.93
$^{230}_{90}\text{Th}$	53.22	174.11	677.51	769.6	1125.6	781.37	775.5
$^{232}_{90}\text{Th}$	49.36	162.12	774.15	873	1078.6	758.25	890.1
$^{234}_{92}\text{U}$	43.49	143.35	851.74	947.64	1044.53	926.72	1023.77
$^{238}_{92}\text{U}$	44.91	148.38	966.13	1056.38	997.23	1037.25	1130.75
$^{238}_{94}\text{Pu}$	44.06	145.93	983.09	1125.75	1134	1028.53	—

3. Conclusions

Regular nuclei have previously been studied using energy levels and quadrupole transition probabilities. The results of the studies indicate the existence of a repetition pattern for the mentioned data. In this article, we studied the existence of repetition patterns and the behavior of regular nuclei using the effective quadrupole deformation of different levels. We calculated the values of the effective quadrupole deformation using IBM. We defined different ratios for the obtained values for different levels of ground, beta, and gamma bands. The results showed that most of the ratios are in the same range (this repetition pattern was observed for the levels of the beta and gamma bands more than the ground band). Therefore, we conclude that there is a direct relationship between the similarity of the deformation values of the nuclei levels and their regularity. Using the random matrix theory (RMT), we also statistically studied the effective quadrupole deformation values. The results obtained from RMT were similar to the behavior of repetition patterns and confirmed them.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Author contributions

A. Hosseinnézhad and H. Sabri performed the initial calculations, analyzed and interpreted the results, and wrote the main manuscript text. All authors commented on and reviewed the manuscript.

Competing interests

The authors declare no competing interests.

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References

- [1] Paar V and Vorkapić D 1990 *Phys. Rev. C* **41** 2397
- [2] Hosseinnézhad A, Seidi M and Sabri H 2023 *Sci. Rep.* **13** 3849
- [3] Porter C E 1965 *Statistical Theories of Spectra: Fluctuations*
- [4] Bonatsos D, Martinou A, Sarantopoulou S, Assimakis I E, Peroulis S and Minkov N 2020 *Eur. Phys. J. Spec. Top.* **229** 2367
- [5] Dong W-T, Zhang Y, He B-C, Pan F, Luo Y-A, Draayer J and Karampagia S 2021 *J. Phys. G: Nucl. Part. Phys.* **48** 045103
- [6] Karampagia S, Zelevinsky V and Spitler J 2022 *Nucl. Phys. A* **1023** 122453
- [7] Bonatsos D, McCutchan E and Casten R 2010 *Phys. Rev. Lett.* **104** 022502
- [8] Hosseinnézhad A, Sabri H and Seidi M 2022 *Nucl. Phys. A* **1022** 122431
- [9] Alhassid Y and Whelan N 1991 *Phys. Rev. Lett.* **67** 816
- [10] Karampagia S, Bonatsos D and Casten R 2015 *Phys. Rev. C* **91** 054325
- [11] Iachello F and Arima A 1987 *Rev. Mod. Phys.* **59** 339
- [12] Mehta M 1991 *Random Matrices* 2nd ed. (Academic)
- [13] Amon L and Casten R 2007 *Phys. Rev. C* **75** 037301
- [14] Amon L and Casten R 2007 *AIP Conf. Proc.* (American Institute of Physics) 91
- [15] Robledo L M and Bertsch G 2012 *Phys. Rev. C* **86** 054306
- [16] Tanabe K and Sugawara-Tanabe K 2008 *Phys. Rev. C* **77** 064318
- [17] Dönauf F 1987 *Nucl. Phys. A* **471** 469
- [18] Wimmer K *et al* 2021 *Phys. Rev. Lett.* **126** 072501
- [19] Kumar K 1972 *Phys. Rev. Lett.* **28** 249
- [20] Cline D 1986 *Annu. Rev. Nucl. Part. Sci.* **36** 683
- [21] Wu C *et al* 1996 *Nucl. Phys. A* **607** 178
- [22] Hosseinnézhad A, Majarshin A J, Luo Y A, Ahmadian D and Sabri H 2022 *Nucl. Phys. A* **1028** 122523
- [23] Hosseinnézhad A, Majarshin A J and Sabri H 2022 *Int. J. Mod. Phys. E* **31** 2250046
- [24] Yeter-Aydeniz K, Pooser R C and Siopsis G 2020 *npj Quantum Information* **6** 1
- [25] Kassim H H, Mohammed-Ali A A, Al-Jubbori M A, Sharrad F I, Ahmed A and Hossain I 2018 46 (1) *Journal of the National Science Foundation of Sri Lanka*
- [26] Jalili Majarshin A, Sabri H and Jafarizadeh M 2018 *Eur. Phys. J. A* **54** 1
- [27] Nomura K, Rodríguez-Guzmán R and Robledo L 2021 *Phys. Rev. C* **104** 054320
- [28] Müller-Gatermann C *et al* 2018 *Phys. Rev. C* **97** 024336
- [29] Greiner W and Maruhn J A 1996 *Nuclear Models* (Springer)
- [30] Arima A and Iachello F 1978 *Ann. Phys.* **111** 201
- [31] Arima A and Iachello F 1987 *The Interacting Boson Model* (Cambridge Univ. Press)
- [32] Arima A and Iachello F 2000 *Ann. Phys.* **281** 2
- [33] Scholten O, Iachello F and Arima A 1978 *Ann. Phys.* **115** 325
- [34] Nomura K and Jolie J 2018 *Phys. Rev. C* **98** 024303
- [35] Macek M, Stránský P, Leviatan A and Cejnar P 2019 *Phys. Rev. C* **99** 064323
- [36] Vallejos O and Barea J 2021 *Phys. Rev. C* **104** 014308
- [37] Poursharif F, Sabri H and Seidi M 2021 *Eur. Phys. J. Plus* **136** 1149
- [38] Nomura K 2022 *Phys. Rev. C* **105** 044301
- [39] Kota V and Sahu R 2021 *Nucl. Phys. A* **1016** 122313
- [40] Guzmán V M B, Flores-Mendieta R and Hernández J 2022 *Eur. Phys. J. A* **58** 1
- [41] Nomura K, Vretenar D, Li Z and Xiang J 2021 *Phys. Rev. C* **103** 054322
- [42] Nomura K, Vretenar D, Li Z and Xiang J 2021 *Phys. Rev. C* **104** 024323
- [43] Sahu R, Kota V and Srivastava P 2020 *Eur. Phys. J. Spec. Top.* **229** 2389
- [44] Cejnar P and Jolie J 2009 *Prog. Part. Nucl. Phys.* **62** 210
- [45] García-Ramos J E and Heyde K 2015 *Phys. Rev. C* **92** 034309
- [46] Iachello F, Zamfir N and Casten R 1998 *Phys. Rev. Lett.* **81** 1191
- [47] García-Ramos J E and Heyde K 2014 *Phys. Rev. C* **89** 014306
- [48] Büyükkata M, Van Isacker P and Uluer I 2010 *J. Phys. G: Nucl. Part. Phys.* **37** 105102
- [49] Cejnar P, Jolie J and Casten R F 2010 *Rev. Mod. Phys.* **82** 2155
- [50] Roca-Maza X and Paar N 2018 *Prog. Part. Nucl. Phys.* **101** 96
- [51] Launey K, Draayer J, Dytrych T, Sun G-H and Dong S-H 2015 *Int. J. Mod. Phys. E* **24** 1530005

- [52] Cseh J, Riczu G and Darai J 2019 *Phys. Lett. B* **795** 160
- [53] Draayer J, Popa G, Hirsch J and Vargas C 2004 *Symmetries In Nuclear Structure: An Occasion to Celebrate the 60th Birthday of Francesco Iachello* (World Scientific) 42
- [54] Heyde K 1992 *Nuclear Shapes and Nuclear Structure at Low Excitation Energies* (Springer) 39
- [55] Van Isacker P and Puddu G 1980 *Nucl. Phys. A* **348** 125
- [56] Werner V, Williams E, Casperson R, Casten R, Scholl C and Von Brentano P 2008 *Phys. Rev. C* **78** 051303
- [57] Leviatan A 1996 *Phys. Rev. Lett.* **77** 818
- [58] Leviatan A 2011 *Prog. Part. Nucl. Phys.* **66** 93
- [59] Nomura K, Gavrielov N and Leviatan A 2021 *Phys. Rev. C* **104** 044317
- [60] Sabri H, Mobarakeh S M, Majarshin A, Luo Y-A and Pan F 2021 *Sci. Rep.* **11** 16417
- [61] Bonatsos D, McCutchan E, Minkov N, Casten R, Yotov P, Lenis D, Petrellis D and Yigitoglu I 2007 *Phys. Rev. C* **76** 064312
- [62] Balankin A S, Martinez-Cruz M and Susarrey-Huerta O 2023 *Modern Physics Letters B* **37** 2250220
- [63] Bialous M and Sirko L 2022 *Phys. Rev. E* **106** 064208
- [64] Sakhr J and Nieminen J M 2005 *Phys. Rev. E* **72** 045204
- [65] Sabri H, Rashidian Maleki B, Fathi H and Jafarizadeh M 2014 *The European Physical Journal Plus* **129** 52
- [66] Alhassid Y and Novoselsky A 1992 *Phys. Rev. C* **45** 1677
- [67] Majarshin A J, Pan F, Sabri H and Draayer J P 2019 *Ann. Phys.* **407** 250
- [68] Majarshin A, Luo Y-A, Pan F, Sabri H, Rezaei M and Draayer J P 2021 *Phys. Rev. C* **104** 024332
- [69] Trallero-Herrero C, Trallero-Giner C, Ulloa S and Perez-Alvarez R 2001 *Phys. Rev. E* **64** 056237
- [70] Zyriliou A, Mertzikis T, Chalil A, Vasileiou P, Mavrommatis E, Bonatsos D, Martinou A, Peroulis S and Minkov N 2022 *The European Physical Journal Plus* **137** 352
- [71] Bonatsos D, Assimakis I, Minkov N, Martinou A, Cakirli R, Casten R and Blaum K 2017 *Phys. Rev. C* **95** 064325
- [72] Davydov A and Filippov G 1958 *Nucl. Phys.* **8** 237
- [73] Esser L, Neuneyer U, Casten R and Von Brentano P 1997 *Phys. Rev. C* **55** 206
- [74] National Nuclear Data Center (Brookhaven National Laboratory), chart of nuclides <http://nndc.bnl.gov>
- [75] Tsunoda Y and Otsuka T 2021 *Phys. Rev. C* **103** L021303
- [76] Otsuka T, Tsunoda Y, Utsuno Y, Shimizu N, Abe T and Ueno H 2023 arXiv:2303.11299
- [77] Rouoof S, Nazir N, Jehangir S, Bhat G, Sheikh J, Rather N and Frauendorf S 2024 *Eur. Phys. J. A* **60** 40
- [78] Sabri H and Majarshin A J 2018 *Nucl. Phys. A* **970** 29
- [79] Möller P, Sierk A J, Ichikawa T and Sagawa H 2016 *Atomic Data and Nuclear Data Tables* **109** 1